PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Risk prediction models for emergence delirium in pediatric general anesthesia: a systematic review
AUTHORS	Petre, Maria-Alexandra; Saha, Bibek; Kasuya, Shugo; Englesakis, Marina; Gai, Nan; Peliowski, Arie; Aoyama, Kazuyoshi

VERSION 1 – REVIEW

REVIEWER	Roger Esmel-Vilomara Vall d'Hebron University Hospital, Barcelona. Spain
REVIEW RETURNED	27-Aug-2020

GENERAL COMMENTS	Although the methodology of the study is impeccable and the review is exhaustive, a review that includes only one article is not enough to draw conclusions. Perhaps it would be useful to propose a prospective observational study with these objectives instead of a review.

REVIEWER	Dr Sarah White St Georges University of London, UK
REVIEW RETURNED	15-Sep-2020

GENERAL COMMENTS	This is a clearly written systematic review with a focused research question that has been conducted using appropriate tools and reporting standards. I applaud the authors for their clarity.

REVIEWER	Dr Anju Gupta All India Institute of Medical Sciences, Delhi
REVIEW RETURNED	13-Oct-2020

GENERAL COMMENTS	The study concept and methodology are sound and the analysis is well executed. The authors should address the following: 1. mention the limitation of their analysis 2. discuss the probable reason why such risk scores are lacking and are important to develop for pediatric anaesthesia 3. Is there a possibility of unpublished studies or some studies after the last date of inclusion of the trials i.e., May 2019, to be missed in the analyisis. 4. The magnitude of problem i.e., emergence delirium and its significance in postoperative phase and the value of valid risk prediction tools needs to be highlighted in discussion. 5. In discussion, there is need to highlight the importance of study findings for practitioners and future researchers as effectively it is an analyis of a single study and the main emphasis is on the lack of studies for prediction

REVIEWER	Andi Ade Wijaya Ramlan Universitas Indonesia, Indonesia
REVIEW RETURNED	26-Oct-2020

GENERAL COMMENTS	This review gives good information about the prediction models for the development of ED in the pediatric population. However, the lack of literature that meets the author's criteria decreased the applicability and acceptability of such models into clinical practice.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1:

Although the methodology of the study is impeccable and the review is exhaustive, a review that includes only one article is not enough to draw conclusions. Perhaps it would be useful to propose a prospective observational study with these objectives instead of a review.

> We thank the reviewer for his comments on the robustness of our methodology and we agree on the need for future research in this area. We have restructured the discussion of the paper to suggest several avenues for future research, beginning with a systematic review of all the patient, surgical and anesthetic factors that predispose children to the development of emergence delirium. We then go on to suggest that these factors could be incorporated in future predictive scales and extensively discuss the performance characteristics which must be included in the validation of such scales in order to ensure their usability in clinical practice. Thank you again for your time and understanding.

Reviewer 2:

This is a clearly written systematic review with a focused research question that has been conducted using appropriate tools and reporting standards. I applaud the authors for their clarity.

> We thank the reviewer for her comments on the robustness of our methodology.

Reviewer 3:

The study concept and methodology are sound and the analysis is well executed. The authors should address the following:

1. mention the limitation of their analysis

> We discuss the major limitation of the analysis which includes that only one study was found, which hampered our ability to run a meta-analysis on the predictive properties of the scale.

2. discuss the probable reason why such risk scores are lacking and are important to develop for pediatric anaesthesia

> We have included in the discussion a section on the difficulties in diagnosis of emergence delirium (ranging from confounding factors and the use of different diagnostic scales for emergence delirium) as possible causes for the paucity of data. We also emphasize why such prediction scales are important to develop for pediatric anesthesia. The included text is as follows:

"Literature in this domain may have been previously lacking due to considerable variations in the definition of emergence delirium, with different diagnostic scales being used, although the PAED scale is the only validated scale to diagnose ED in children at the moment.[16] Furthermore, several confounding factors are present in the postoperative period, which makes timely and accurate diagnosis of pediatric ED difficult in clinical and also research settings.[1-3] As such, strategic prevention of pediatric ED based on a precise and validated prediction scale is preferable to treatment when the event occurs."

3. Is there a possibility of unpublished studies or some studies after the last date of inclusion of the trials i.e., May 2019, to be missed in the analysis.

The literature search has been updated to November 17, 2020.

4. The magnitude of problem i.e., emergence delirium and its significance in postoperative phase and the value of valid risk prediction tools needs to be highlighted in discussion.

Thank you for your suggestion. We have highlighted these points throughout the discussion. Several sentences in the Discussion have been added, as follow:

"The current systematic review revealed only one scale targeted at predicting the risk of developing emergence delirium in children, a complication associated with significant morbidity occurring in approximately 25% of pediatric general anesthetics."

"Literature in this domain may have been previously lacking due to considerable variations in the definition of emergence delirium, with different diagnostic scales being used, although the PAED scale is the only validated scale to diagnose ED in children at the moment.[16] Furthermore, several confounding factors are present in the postoperative period, which makes timely and accurate diagnosis of pediatric ED difficult in clinical and also research settings.[1-3] As such, strategic prevention of pediatric ED based on a precise and validated prediction scale is preferable to treatment when the event occurs."

"Lastly, future research, quality improvement projects and cost-benefit analyses should focus on determining whether implementing such scales into clinical practice to help target prophylaxis and treatment of emergence delirium results in improved patient outcomes with adequate applicability and acceptability of such scales into clinical practice."

5. In discussion, there is need to highlight the importance of study findings for practitioners and future researchers as effectively it is an analysis of a single study and the main emphasis is on the lack of studies for prediction

We have expanded the discussion about the potential research implications for future research.

Thank you very much for your time and comments, which improves our paper significantly.

Reviewer 4:

This review gives good information about the prediction models for the development of ED in the pediatric population. However, the lack of literature that meets the author's criteria decreased the applicability and acceptability of such models into clinical practice.

We agree with the reviewer about the findings of the study. The strength of the current paper is to highlight this paucity of data, to suggest the next steps to be taken to remedy this, as well as means of structuring future research to make sure the results are robust and usable in clinical practice. Thank you again for your time and comments, appreciated.

VERSION 2 – REVIEW

REVIEWER	Dr Anju Gupta All India Institute of Medical Sciences, New Delhi-110076
REVIEW RETURNED	08-Dec-2020

GENERAL COMMENTS	The concerns have been appropriately addressed by the reviewers